

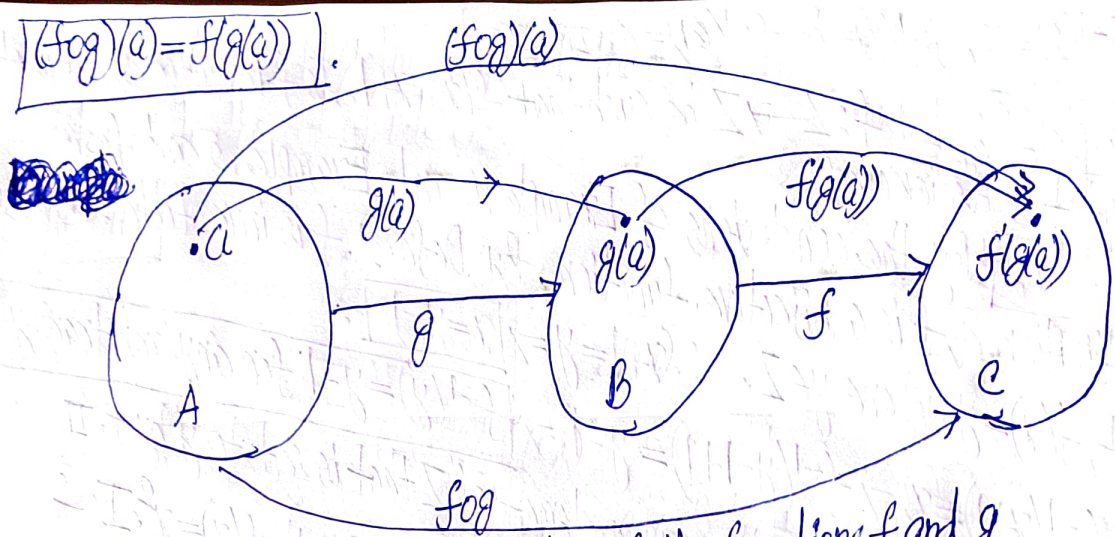


FIGURE 7 The composition of the functions f and g

In other words, $f \circ g$ is the function that assigns to the element a of A the element assigned by f to $g(a)$. The domain of $f \circ g$ is the domain of g .

The range of $f \circ g$ is the image of the range of g with respect to the function f . That is, to find $(f \circ g)(a)$ we first apply the function g to a to obtain $g(a)$ and then we apply the function f to the result $g(a)$ to obtain $(f \circ g)(a) = f(g(a))$.

• Note that the composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f .

Example 23: Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$ and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution: $f(g(a)) = f(b) = 2$ $f(g(b)) = f(c) = 1$ $f(g(c)) = f(a) = 3$.

$g \circ f$ is not defined because the range of f is not a subset of the domain of g . $\square$

Example 24: Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?